# HVAC Troubleshooter

Field-Based Decision Trees

**Purpose.** Convert documented troubleshooting cases into repeatable diagnostic workflows for an offline commercial hydronic HVAC reference. These are preliminary field guides, not manufacturer-approved procedures. Verify model-specific meanings, limits, terminals, and tests against the exact manual. Follow lockout/tagout and electrical safety requirements; never bypass a safety device.

## 1. Tekmar 256 / 274 — Outdoor Temperature Is Incorrect

Related cases: Tekmar 256 reading changed from about 65°F to 82–85°F; Tekmar 274 showed about 62°F when outdoor temperature was about 54°F.

### *START*

Controller outdoor temperature is missing, unstable, or does not match a reliable independent measurement.

### *A*

Identify the exact controller model and the configured outdoor-temperature source. Is it a physical sensor, network value, or Internet/weather source supported by this model?

### *A → physical sensor*

Inspect sensor wiring and connections. Compare sensor resistance at the measured sensor temperature with the correct manufacturer resistance chart. Isolate the circuit as directed before resistance testing.

### *If resistance matches chart*

Check sensor placement, exposure to weather, configuration, and whether the controller is displaying the intended input. Continue to manufacturer-specific input diagnostics if the value remains wrong.

### *If resistance does not match chart*

Check for loose/corroded connections, damaged cable, moisture, short/open circuit, or a failed sensor. Repair/replace only as supported by test results and manufacturer procedure.

### *A → network / Internet source*

Verify source selection, network connectivity, weather-data availability, date/time configuration, and any required network setup. Do not diagnose a physical sensor when the controller is not using it.

### *If value jumps suddenly*

Prioritize intermittent wiring/connection faults or unstable source data. Record values and timestamps; avoid changing multiple settings at once.

### *VERIFY*

Confirm the reading is stable and plausible, then review outdoor reset and WWSD settings. Verify boiler response using the approved test procedure.

**Field caution:** Important: the reported UP-button behavior on the Tekmar 256 (flame indication and burner running only while the button was held) requires confirmation in the exact manual. Do not assume this is normal or use it as a substitute for a proper functional test.

## 2. Tekmar 671 — “Outdoor Sensor Open Circuit” but Outdoor Source Is Internet

Related case: Tekmar 671 showed an outdoor-sensor open-circuit alarm while Outdoor Sensor was set to Internet; time was missing until changed to AUTO, after which the alarm cleared and an update started.

### *START*

Tekmar 671 reports an outdoor sensor open-circuit alarm.

### *1*

Record the exact alarm text and current configuration. Confirm the Outdoor Sensor source setting and whether the installation is intended to use a physical sensor or Internet weather data.

### *2A — physical sensor intended*

Check the correct sensor terminals, cable, connections, and sensor resistance using the exact manual. Do not infer terminal assignments from wire colors alone.

### *2B — Internet source intended*

Check network connection, IP/network configuration, date/time, source settings, and whether weather data is being received. Check whether the displayed outdoor/weather fields contain valid values or dashes.

### *3*

If date/time is missing or incorrect, follow the manual’s time configuration procedure. If AUTO is an available documented setting, record the original setting before changing it.

### *4*

After any correction, confirm the alarm status, current outdoor temperature, weather forecast data, and update status. Allow the documented update interval if applicable.

### *VERIFY*

Confirm the error remains cleared and the intended outdoor data source is updating. Record the exact change that preceded recovery.

**Field caution:** Observed correlation is not proof of root cause: in the prior case, changing the time setting to AUTO coincided with the alarm clearing and an update starting. Do not generalize this into a universal fix.

# 3. B&G; Ecocirc — E02 High Motor Current / Fault Returns After Reset

Related case: pump displayed E02; power was cycled, the pump ran through startup, then faulted again.

### *START*

Ecocirc pump stops with E02 or returns to E02 after a restart.

### *1*

Record the complete model, fault, operating condition, and whether the fault appears immediately at startup or after running. Preserve any other diagnostic indications.

### *2*

Follow the manufacturer’s safe shutdown and isolation procedure before inspecting the pump. Do not repeatedly power-cycle a persistent fault.

### *3*

Check installation and operating conditions against the manual. Look for evidence of mechanical binding, obstruction, abnormal noise, or damage only where safely accessible and permitted.

### *4*

Have a qualified person verify supply conditions and wiring against the unit nameplate and manufacturer limits. Do not assume the motor itself is the only possible cause.

### *5A — visible/mechanical issue found*

Correct the issue only according to the approved procedure, then retest.

### *5B — no issue found or fault persists*

Follow the exact model’s diagnostic procedure and manufacturer service/replacement criteria. Escalate rather than guessing at internal failure.

### *VERIFY*

After correction, confirm the pump completes startup and operates under normal conditions without the fault recurring. Record measurements and duration of the test.

**Field caution:** Field note: a recurring E02 after power cycling was treated in the prior case as a persistent high-motor-current problem, with pump/motor failure among the possibilities. The fault code and replacement decision should be verified against the exact Ecocirc model manual.

# 4. B&G; Ecocirc — E06 / Suspected Incoming Voltage Imbalance

Related case: two Ecocirc pumps were reported to stop and require manual resets; E06 was described as incoming voltage imbalance outside the pump's ±10% range.

### *START*

Pump stops intermittently or records E06.

### *1*

Record exact pump model, fault history, supply type, and whether the problem correlates with generator operation, a building shutdown, or other electrical events.

### *2*

A qualified person measures the incoming supply and checks connections using appropriate electrical safety procedures. Compare results with the exact pump manual; confirm whether the published limit refers to imbalance, deviation, or another specified measurement.

### *3A — readings outside specification*

Investigate source quality, phase/line conditions as applicable, loose connections, and upstream electrical equipment. Correct the electrical issue before returning the pump to service.

### *3B — readings within specification during test*

Intermittent problems may not be present during a single measurement. Review event timing, supply records, connections, and manufacturer diagnostics. Do not conclude the pump is healthy from one normal reading.

### *4*

Review the pump's restart and reset behavior per the manual. Do not repeatedly reset without investigating the supply issue.

### *VERIFY*

Monitor through a representative operating period and, if relevant, through the operating condition associated with the original fault.

**Field caution:** Safety: electrical supply testing is for qualified personnel using appropriate PPE and procedures. Do not open energized equipment unless the task and risk controls specifically permit it.

# 5. Danfoss VLT HVAC Drive — Unknown “FA04” Setting or Parameter

Related case: customer asked how to enter an FA04 setting on a makeup skid; exact drive model and meaning of FA04 were not yet confirmed.

### *START*

A request references “FA04,” but the correct programming action is unclear.

### *1*

Read the drive nameplate and capture the exact model/product family (for example, VLT HVAC Drive FC 102), voltage, and application.

### *2*

Clarify where FA04 appears: on the drive display, in a parameter list, in a fault/alarm record, or in skid documentation. Obtain a photo or exact text if possible.

### *3A — parameter number confirmed*

Open the manual for that exact drive and firmware/application. Confirm the parameter name, units, permitted range, access level, and effect before editing.

### *3B — fault/alarm code or document reference*

Do not treat it as a parameter. Look up the exact code/reference in the correct drive and skid documentation.

### *4*

Before changing a pressure-related setting, record current values and confirm the intended pressure setpoint with the responsible engineer/customer. Check transducer range/scaling and system design where relevant.

### *5*

Apply only an authorized, documented change. Record old value, new value, reason, and who approved it.

### *VERIFY*

Confirm the drive starts and controls the intended process correctly, with no new alarms and with the pressure response matching the approved setpoint.

**Field caution:** Do not guess what “FA04” means from the identifier alone. Do not lower pressure settings until the required setpoint and consequences are confirmed.

# 6. AERCO BME — External Interlock Alarm Despite Apparent Continuity

Related cases: BME Boiler 2A external interlock alarm despite reported continuity/voltage checks; another service note describes backing up controller parameters, resetting/restoring them, and alarms clearing afterward.

### *START*

AERCO BME reports an external interlock alarm.

### *1*

Record exact alarm wording, boiler tag, controller revision, operating state, BMS command/state, and whether the alarm is active or historical.

### *2*

Use the unit-specific wiring diagram and manufacturer procedure to identify the interlock circuit and required state. Do not infer terminals or acceptable voltage from another model.

### *3*

A qualified person checks the prescribed circuit state, wiring, terminals, external contacts, and BMS logic. Continuity alone may not prove that a circuit behaves correctly under operating conditions.

### *4A — interlock state is not as required*

Trace the external circuit and BMS sequence to locate the cause. Repair the actual issue; never jumper or bypass the interlock to force operation.

### *4B — circuit appears correct*

Verify the controller configuration, alarm history, and related permissives against the manual. Check whether the signal changes correctly during the approved functional test.

### *5*

Only if justified and authorized, back up the current parameter set before any reset/restoration. Follow AERCO's exact procedure and restore the approved configuration.

### *VERIFY*

Confirm the interlock responds correctly, the alarm clears by the prescribed method, and the boiler operates normally with all safety devices functional.

**Field caution:** Field note: a prior case reported that backing up the program and resetting/restoring parameters was followed by the alarms clearing. This is an observed sequence, not evidence that a programming reset is the correct first response to every interlock alarm.

# 7. Hydronic Pump — Flow/Heat Delivery Complaint After Startup

A general triage tree for pump/startup calls. Use the actual design and pump documentation; do not assume a universal target delta-T or flow rate.

### *START*

A hydronic system reports poor heat delivery, slow recovery, or a pump that appears to run without achieving the expected result.

### *1*

Confirm the complaint and system conditions: which loop/equipment, load state, valve positions, operating schedule, and whether the problem began after service or a shutdown.

### *2*

Confirm the pump is commanded to run and is actually running. Check displayed status, active alarms, drive command, and any BAS enable/interlock status.

### *3*

Verify system valves and isolation valves are in their intended positions, and check for obvious air, leakage, blockage, or strainers requiring attention under the system procedure.

### *4*

Record supply/return temperatures and calculate delta-T. Compare with the project design, commissioning documents, or equipment manual—not a universal assumed range.

### *5*

If the pump uses a VFD, verify the approved command/reference, actual speed/frequency, motor data, and sensor/transducer configuration. Do not raise speed or change setpoints without confirming design limits.

### *6*

If temperatures/flow remain abnormal, investigate balancing, air removal, control-valve operation, heat-exchanger fouling, and system-side restrictions in a controlled order.

### *VERIFY*

Confirm stable operation at representative load, expected temperatures/flow per design, no new alarms, and no leaks or abnormal noise.

**Field caution:** Use this as a generic tree only. A pump running does not prove that required flow is present; likewise, delta-T alone does not identify the root cause.